

Integrated Optical and Electrical Analysis: Identifying Location and Properties of Traps in AlGaIn/GaN HEMTs During Electrical Stress

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Abstract—A new methodology is developed to determine spatial location and properties of traps generated by electrical stressing of AlGaIn/GaN high-electron mobility transistors, based on integrated optical and electrical analysis. Mild OFF-state stress increases irreversibly the number of traps located in the near-surface AlGaIn region at the gate edge. A deep level with 0.45-eV activation energy in fresh devices changes its nature to interacting defect after the OFF-state stress, accompanied by an activation energy change. These results are consistent with trap generation in the near-surface AlGaIn region at the gate edge related to high electric field and gate leakage current, as stressing does not result in the generation of cracks in the AlGaIn layer.

Index Terms—Electrical stress, electroluminescence (EL), GaN high-electron mobility transistor (HEMT), impurity diffusion, reliability, trapping.

ALTHOUGH AlGaIn/GaN high-electron mobility transistors (HEMTs) have achieved impressive power RF performances, they still suffer from long-term reliability issues, such as increased carrier trapping [1] and related RF power density drop after stressing [2], [3]. Degradation mechanisms have been attributed to hot-electron induced trap generation [1], converse piezoelectric effect [4], [5], or deep-trap-level generation [6]. However, there is still only a very limited understanding of degradation mechanisms in GaN HEMTs, mostly due to the fact that common dc and pulsed current–voltage (I – V) analysis is insufficient to discriminate between defect evolution in different transistor regions, i.e., there is no direct information on the location of traps generated in a device, as well as the nature of the traps.

In this letter, we report on the development of a new methodology to study the degradation mechanisms in AlGaIn/GaN HEMTs by using a unique combination of optical and electrical

techniques: Raman thermography, electroluminescence (EL), and trapping analysis. An irreversible increase in traps located in the AlGaIn barrier close to the drain side of the gate edge was observed, together with a change in their activation energy.

AlGaIn/GaN HEMTs were grown on SiC substrates using metal–organic chemical vapor deposition, with 25-nm-thick $\text{Al}_{0.26}\text{Ga}_{0.74}\text{N}$ on a 2- μm semi-insulating GaN buffer and a 0.7-nm AlN layer in between. Standard Ti/Al/Ni/Au source/drain ohmic metallization annealed at 870 °C for 30 s and a rectangular-shaped Ni/Au gate contact were used. Devices were passivated using 160-nm SiN_x deposited by plasma-enhanced CVD. HEMTs with source–gate spacing, gate–drain spacing, gate length, and gate width of 1, 2.5, 0.7, and 150 μm , respectively, were subjected to OFF- and ON-state stressings. To determine trap location and properties, integrated optical and electrical device analysis was performed. EL spatial and spectral imaging integrated into Raman thermography determined information on device temperature, hot-electron temperature distribution, and electric-field strength. The shift resolution of EL imaging was 0.1 μm . More details on optical techniques can be found in [7] and [8]. Detrapping transient characteristics were used to determine relative trap densities and activation energies (E_a). Drain current (I_d) transients were measured at $V_{gs} = 1$ V and $V_{ds} = 0.5$ V after a 1-s-long $V_{gs} = -10$ V and $V_{ds} = 0$ V filling pulse similar to [9]. However, a different approach to extract trap time constants was used, providing the same quantitative results. I_d transients were first fitted using a polynomial function $I_d(x) = a_0 + \dots a_{n-1}x^{n-1}$ ($n = 10$; $x = \log(\text{time})$), and then, $dI_d(x)/dx$ was deconvoluted using Gaussian functions. Four Gaussian functions were adequate to fit $dI_d(x)/dx$ with mean, variance, and amplitude as fitting parameters. Gate lag, defined as a difference between dc and pulsed (only V_{gs} pulsed from -5 to 0 V; 100-ns pulsewidth) output I_d , was measured before stress and one day after stress. OFF-state stress was performed in pinchoff with $V_{gs} = -5$ V and $V_{ds} = 30$ V up to 10 h at 25 °C base plate temperature. A selection of devices was studied, with all devices showing the same degradation behavior; the following restricts to the discussion of a representative device. Gate leakage measured at $V_{gs} = -7$ V and $V_{ds} = 20$ V increased progressively from 3 to only 8.6 mA/mm, i.e., mild stress condition. For comparison, ON-state stress was also applied, at $V_{gs} = 0$ V and $V_{ds} = 30$ V for 10 h at 25 °C (channel temperature ~ 300 °C). Gate leakage increased from 2.6 to 6.4 mA/mm. Transmission electron microscopy (TEM) analysis was performed on unstressed and stressed devices.

Manuscript received March 11, 2010; revised March 23, 2010; accepted March 25, 2010. Date of publication May 6, 2010; date of current version June 25, 2010. This work was supported by the Office of Naval Research (ONR) and ONR Global under Grant N00014-08-1-1091 through the DRIFT Program (monitored by Dr. Paul Maki). The review of this letter was arranged by Editor G. Meneghesso.

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Digital Object Identifier 10.1109/LED.2010.2047092

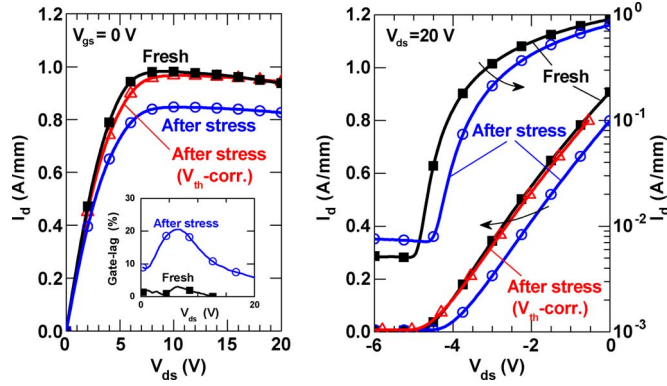


Fig. 1. Output and transfer characteristics of AlGaIn/GaN HEMT before and one day after 10-h OFF-state stressing at $V_{ds} = 30$ V and $V_{gs} = -5$ V. Characteristics corrected for the shift in V_{th} are also shown. Inset shows the gate lag as a function of V_{ds} measured before and one day after OFF-state stress.

OFF-state stress induces a decrease in saturation drain current, mostly due to a positive threshold voltage (V_{th}) shift (Fig. 1) and an increase in gate-lag (inset of Fig. 1). Temperature profiles measured across the source–drain region obtained using Raman thermography showed no significant changes in device temperature distribution before and after OFF-state stress. This is consistent with EL data [Fig. 2(a)]. EL originates from intraband optical transitions of hot carriers in the AlGaIn/GaN device channel [8], showing only a $\sim 30\%$ decrease in EL intensity near the gate contact after stressing. From the dependence of EL intensity on $1/(V_{ds} - V_{ds,sat})$ and electric-field distribution simulations [10], this indicates that the peak electric field decreases by less than 10%. We observe a small shift of the EL peak intensity from the gate edge toward the source after stressing, as shown in Fig. 2(a). EL peak is shifted already after the first hour of OFF-state stress and then remains almost unchanged. This is in contrast to ON-state stressing where no change in EL peak position was observed.

EL spectra were measured at different points across the source–drain region before and after OFF-state stress [Fig. 2(b)]. The high-energy tail of the EL spectrum was used to determine electron temperature (T_e) profile [8] across the transistor channel, shown in Fig. 2(b), together with peak T_e near the gate contact as a function of V_{ds} . After the OFF-state stress, T_e decreased with respect to the fresh device, with its spatial profile showing a distinct change from the gate contact toward the drain. This clearly shows that stressing affects predominantly the gate region. Slight shift in the EL peak position [Fig. 2(a)] indicates that the generated traps are located close to the drain edge of the gate contact. There are two mechanisms for changes in T_e , both related to traps generated during stressing: 1) traps only affecting electric-field distribution and 2) traps also resulting in increased electron scattering [11]. We find a similar slope in the linear relationship between T_e and V_{ds} before and after stressing [inset of Fig. 2(b)], suggesting unchanged carrier-scattering mechanism. No dramatic increase in carrier scattering near the gate indicates that the generated traps are closer to the AlGaIn surface rather than directly in the device channel, which is also consistent with the observed gate-lag increase (inset of Fig. 1).

To determine the trap type generated by stressing near the gate edge, the optical analysis was linked to the electrical characterization of traps. Fig. 3(a) shows the time constant analysis

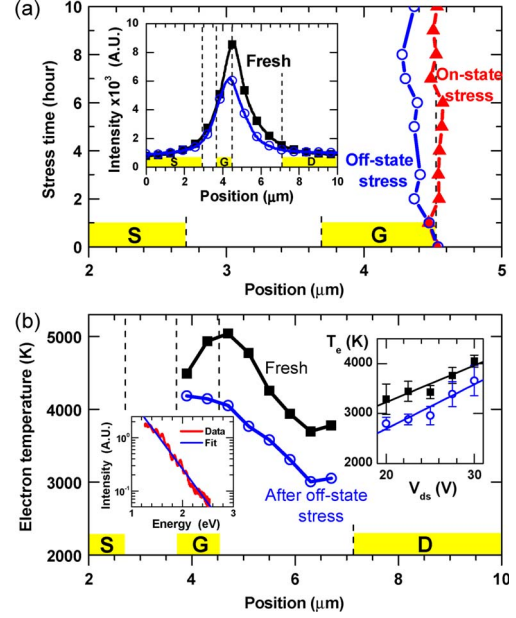


Fig. 2. (a) EL intensity peak position during OFF- and ON-state stressings as a function of stress time. Inset shows the EL distribution before and after 10-h OFF-state stress. (b) Electron temperature (T_e) in AlGaIn/GaN HEMT across the active device region, determined from the high-energy tail of the EL spectra (exemplified in the left inset), before and after OFF-state stress. Right inset shows peak T_e as a function of V_{ds} before and after OFF-state stress. All EL measurements were performed at $V_{ds} = 30$ V and $V_{gs} = 0$ V at the center of the device finger 40 μm away from the mesa edge.

of detrapping transients measured before, during, and after OFF-state stress. Two temperature-dependent trap levels (Tp1 and Tp2) with E_a values of 0.45 and 0.12 eV and one temperature-independent trap (Tp3) were found in the fresh devices. All traps increased in density during the OFF-state stress [Fig. 3(a) inset] with only slight recovery measured after one day. In the following, we focus on the dominant trap Tp1. The temperature dependence of Tp1 time constant after OFF-state stress [inset of Fig. 3(b)] now showed two activation energies ($E_a = 0.05$ and 0.65 eV), indicating Tp1 to evolve into a more complex defect, such as multilevel or multiconfigurational traps [12], denoted as Tp1* in the following. To further study this change, Tp1* was measured as a function of filling pulse duration (t_f) [Fig. 3(b)]. The amplitude of Tp1* after OFF-state stress linearly increases with the logarithm of t_f , in contrast to the fresh device, where no significant change with t_f was found. This behavior suggests emergence of interacting defects as a result of repulsive time-dependent potential of trapped electrons [13]. Compared to the OFF-state stress, only a smaller increase ($\sim 30\%$) in the Tp1 amplitude was observed one day after ON-state stressing, which is consistent with the EL peak position not being affected by stressing [Fig. 2(b)].

It has been suggested that degradation under the OFF-state stress is often related to mechanical damage due to the converse piezoelectric effect [5]. Cross-sectional TEM micrographs of the investigated device after OFF-state stress reveal no mechanical crack formation on the drain side of the gate. Moreover, the observed change in E_a of Tp1 would be too large to be related to strain relaxation expected by crack formation (< 60 meV). This indicates that other mechanisms need to be considered for the early stages of device degradation during OFF-state stress. As plotted in the inset of Fig. 3(a),

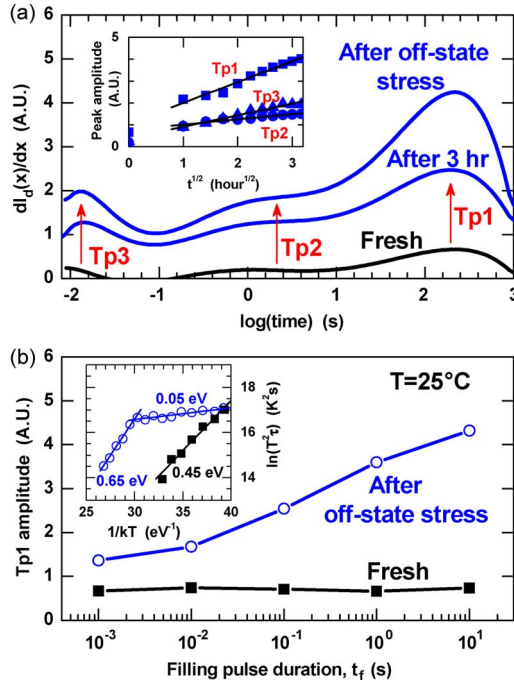


Fig. 3. (a) $dI_d(x)/dx$ versus $\log(\text{time})$ plot of detraping transients measured at $V_{gs} = 1$ V and $V_{ds} = 0.5$ V after 1-s-long $V_{gs} = -10$ V and $V_{ds} = 0$ pulse at 25 °C before, during, and after 10-h OFF-state stress. Inset shows the evolution of trap amplitude as a function of square root of stressing time. (b) (Solid squares) Tp1 and (open circles) Tp1* peak amplitudes as functions of filling pulse duration t_f measured by detraping transient as in (a) and (inset) Arrhenius plot of Tp1 and Tp1* time constant (τ) before and after OFF-state stress.

the trap amplitudes increase with near-square-root dependence on stressing time, indicating a diffusion-related degradation processes. Local gate leakage current, coupled with high fields and converse piezoelectric strain may accelerate diffusion of impurities such as C and O into AlGaIn close to the drain side of the gate, similar to defect migration known for GaAs and GaP [14]. A possibility would be C substituting Ga that was predicted to introduce deep level in GaN located 0.5 eV below the conduction band edge [15], consistent with the Tp1 in fresh device; however, other impurities/defects may also play a role. The higher V_{dg} during OFF-state stress may explain the faster degradation evolution compared to ON-state stress, while reverse voltage applied across the gate–source diode during OFF-state stress can induce larger V_{th} shift. Due to the different position of the Fermi level at the AlGaIn surface below the gate and in the access region, Tp1* traps generated during OFF-state stress underneath the gate can be negatively charged, while those generated in the access region close to the drain side of the gate edge can be charged positively, explaining the positive V_{th} shift with transconductance almost unaffected [Fig. 1(b)] and an effective shift of the EL peak toward the source side [Fig. 2(a)]. Finally, as a result of impurity diffusion into the AlGaIn layer, formation of C- and/or O-related complexes may explain the observed change in properties of Tp1*; however, other mechanisms may also contribute. Such generation of defects could serve as nucleation for crack formation reported for harsher stress conditions [5].

In conclusion, a methodology of integrated optical and electrical analysis has been developed to determine spatial location and properties of traps generated in GaN HEMTs under electrical stress. During OFF-state stress, traps are generated close to the drain side of the gate located in the near-surface region of the AlGaIn layer, likely due to a diffusion process driven by high electric field and gate leakage current. Change of the trap E_a and transition to interacting point defect was found. ON-state stress did not produce any permanent damage.

ACKNOWLEDGMENT

The authors would like to thank J. del Alamo (MIT) for the fruitful discussions and G. D. Via (WPAFB) for the TEM analysis.

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